

Elements of Data Science

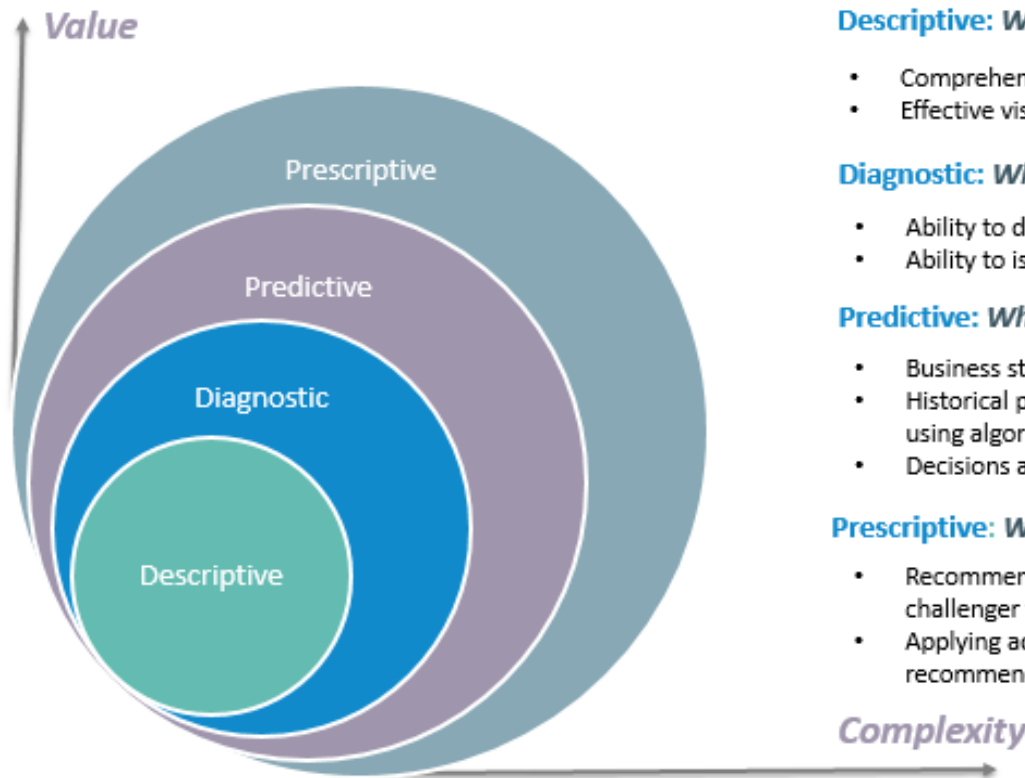
Session I – Introduction

7th September 2019

What is Analytics?

❖ *Analytics is the discovery and communication of meaningful patterns in data to drive smart decisions.*

4 types of Data Analytics



What is the data telling you?

Descriptive: *What's happening in my business?*

- Comprehensive, accurate and live data
- Effective visualisation

Diagnostic: *Why is it happening?*

- Ability to drill down to the root-cause
- Ability to isolate all confounding information

Predictive: *What's likely to happen?*

- Business strategies have remained fairly consistent over time
- Historical patterns being used to predict specific outcomes using algorithms
- Decisions are automated using algorithms and technology

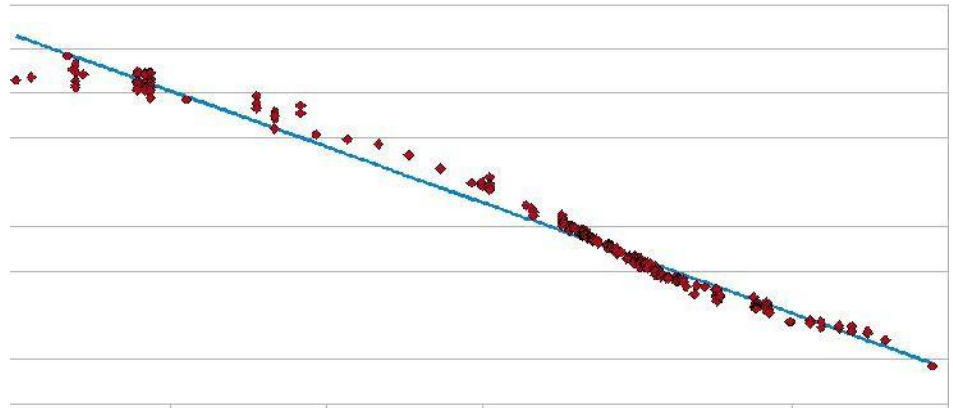
Prescriptive: *What do I need to do?*

- Recommended actions and strategies based on champion / challenger testing strategy outcomes
- Applying advanced analytical techniques to make specific recommendations

Why is it hot?

**Cost
(Per GB)**

1981 — \$300,000
1987 — \$50,000
1990 — \$10,000
1994 — \$1000
1997 — \$100
2000 — \$10
2004 — \$1
2010 — \$0.10



Volume



Every minute:

Google 
2,000,000 searches

facebook
684,578 updates


204,166,667 sent

You Tube 
48 hours of video uploaded²

Business Applications

Credit card companies

- Who should I offer my credit cards to?
- How do I decide on the credit limit for each customer?
- How do I fix the interest rate?
- How do I identify fraud?
- How do I predict bankruptcy?

Retailers

- How do I stock the products in order to maximize my profit?
- How do I market to my customers to maximize my appeal?
- How do I price new existing products in my store?
- How do I design promotion strategies to maximize customer benefit?

FMCG Companies

- How do I price a product?
- How do I optimize inventory to minimize costs?
- How do I determine if a particular ad campaign is working or not?
- What is the return on investment from my marketing budget?

Telecom

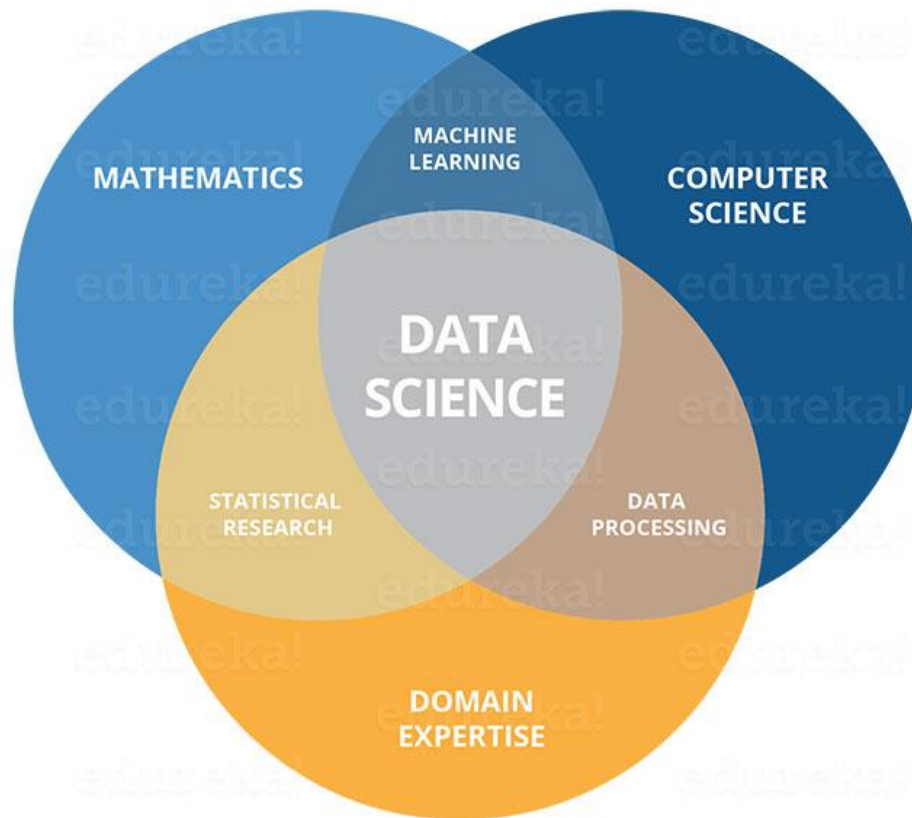
- How do I manage credit limits for my post paid connections?
- How do I sell more value add services?

Insurance

- How to identify early claims?
- Which policies are likely to renew?
- Which customers are likely to surrender policies?
- Cross Sell / Up Sell
- Personalized Marketing

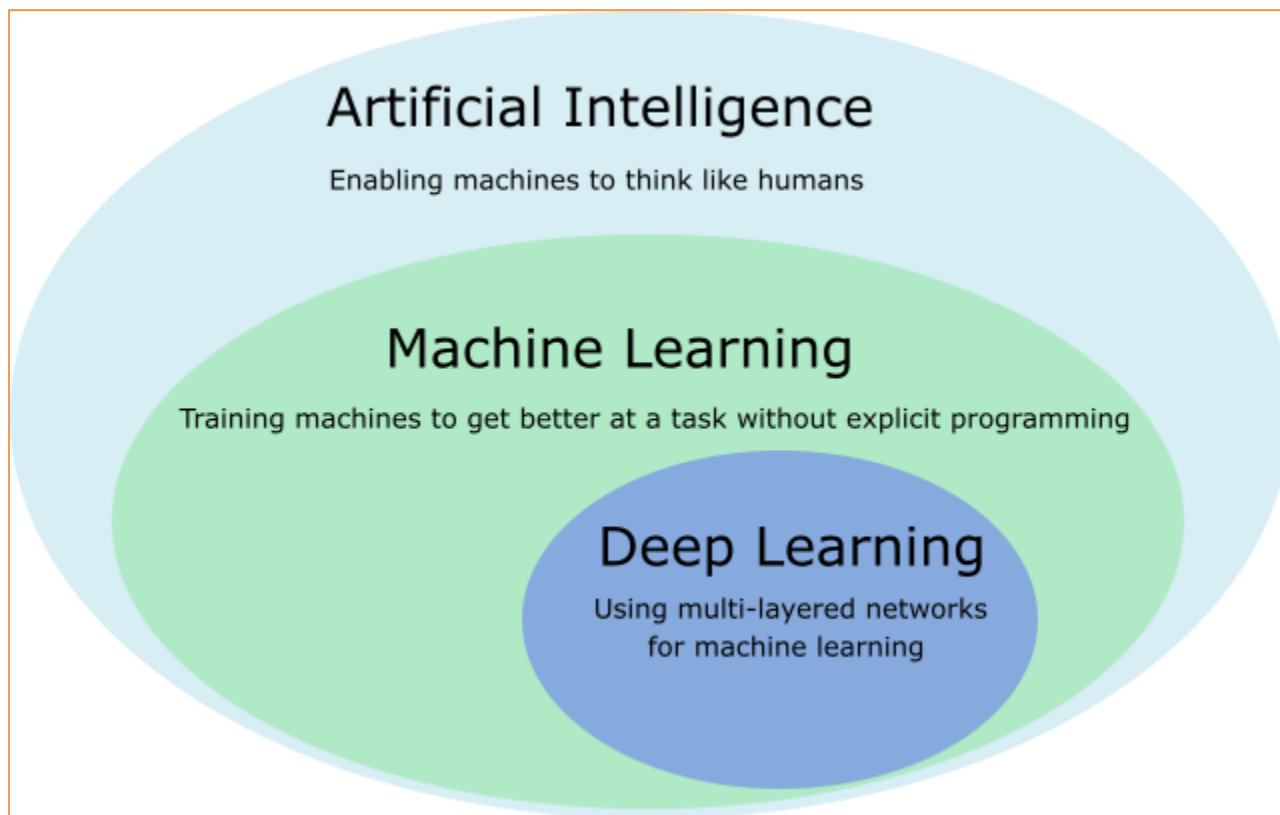
What is Data Science?

- ❖ Data Science is a blend of various tools, algorithms, and machine learning principles with the goal to discover hidden patterns from the raw data.
- ❖ How is this different from what statisticians have been doing for years?
 - The answer lies in the difference between explaining and predicting.

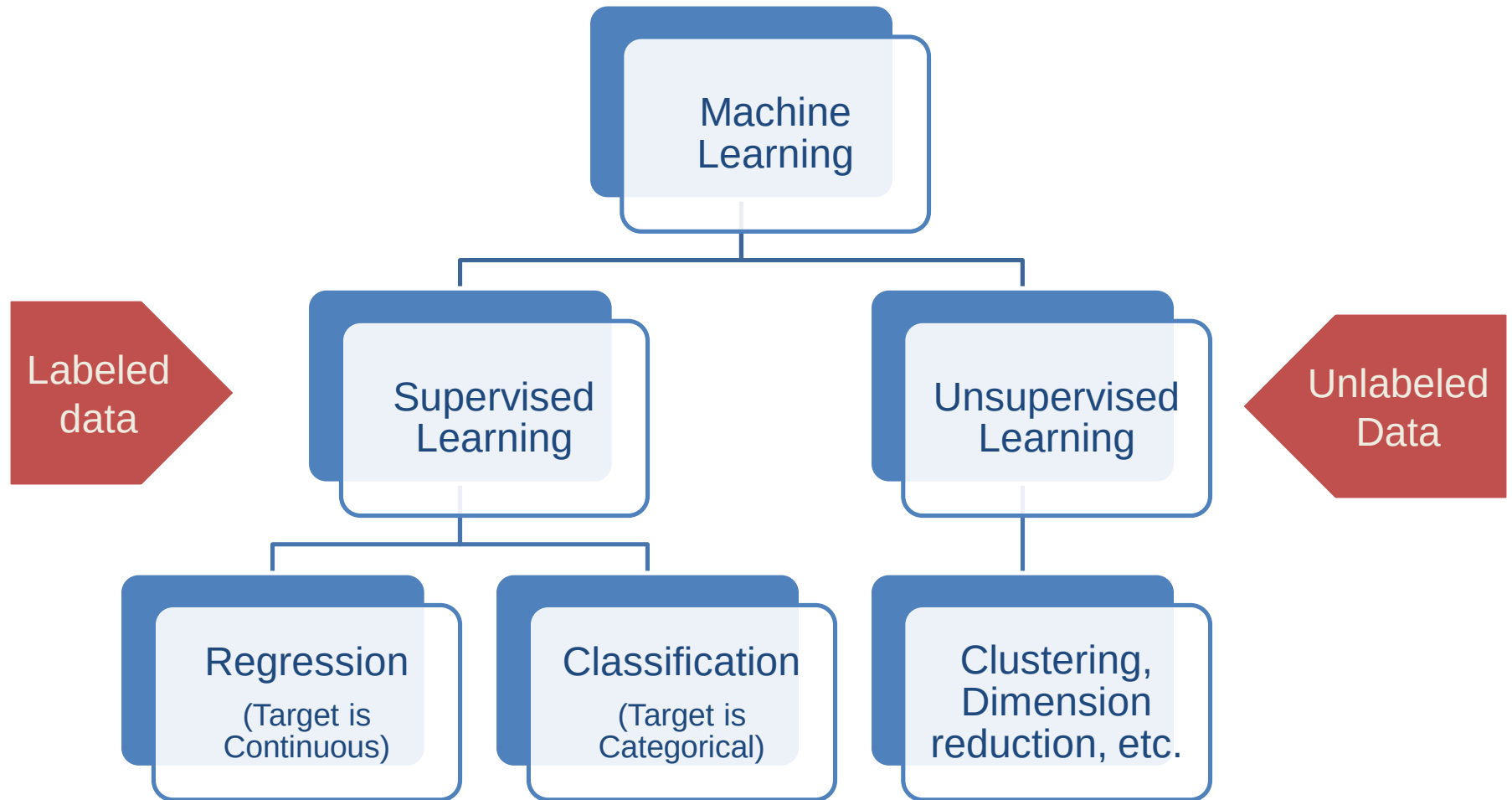


Machine Learning

- ❖ Machine Learning: Field of study that gives computers the ability to learn without being explicitly programmed.
 - Arthur Samuel (1959)



Types of Machine Learning



Stages in analytics project

1

- **Problem Definition :**
 - Understanding business problem and converting it into an analytical (data) problem

2

- **Data Collection:**
 - Understanding the available data, create data repository as per the problem

3

- **Exploratory Data Analysis (EDA)**
 - Understanding variables, Univariate / Bivariate analysis
 - Missing Value/Outlier treatment, Label & One hot Encoding, etc.

4

- **Feature Engineering:**
 - Creating new variables based on business problem (Age of the customer, Customer vintage, # purchases in last 6 months, etc.)

5

- **Data Partition (Sampling):**
 - Creating training, validation, testing, and out of sample data (Based on time periods)

6

- **Model Building:**
 - By selecting appropriate algorithms, building multiple models on training data
 - Selecting the best model based on the validations performed on unseen data

7

- **Implementation:**
 - Insights and Recommendations based on the discussion with the business
 - Creating model scoring scripts (To be used to score whenever new predictions are needed)

Data Preparation

- ❖ Data in the real world is dirty
 - Incomplete / Noisy / Inconsistent
- ❖ Garbage In => Garbage Out
- ❖ What questions should you ask yourself before you get ready to hit the ground running:
 - Do I have all datasets? (Check which databases contain the required tables)
 - Do I understand all the variables? (Create a Data dictionary)
 - What will be the unique key at which the model will be developed?
 - What is the format of these variables? (Special checks for Dates)
 - What is the primary key? (Understand joining keys)
 - Do I know the frequency of refresh? (Periodicity of data)
- ❖ Predictive Model questions:
 - What is the event to be predicted? (Target variable)
 - What will be the universe considered for the model development?
(Check if there are any special conditions)
 - What will be the frequency of predictions?



Data Cleaning

❖ Data problems which requires data cleaning

➤ Duplicate records

- Same policies (unique record) duplicating
 - Check the count for all rows and unique rows
- Someone taking same survey twice

➤ Incomplete data

- Income / Occupation / Education profiles of customers are missing or kept as common
- Inconsistent data : Customer Id Numeric in one dataset while character in another
- Check trailing / leading blanks (specially in keys)

➤ Event (target variable) should have happened before the prediction date

➤ Check if any of the variable is using future information beyond prediction date

Missing Value

❖ Missing data may be due to:

- Data entry problem
- Data Available for limited period
- Technology limitation

❖ How to handle missing values:

- Ignore if large % of observations missing (~75%+)

➤ Categorical variables:

- Impute using common level (Mode)
- Create a new category as “missing” or “others”
- Use any random value to fill in (999999)

➤ Continuous Variables

- Impute with mean / median
- Impute using other variables in data
- Imputation using KNN, etc.

Outlier Treatment

❖ What is an Outlier:

- ❖ “Frequently the most interesting points of a data set are the points that do not seem to belong; i.e., they seem to differ by a substantial amount from the rest of the data. We call these points outlier”

(Check if any variable contains infinite values – specially ratios)

❖ How to handle such Values?

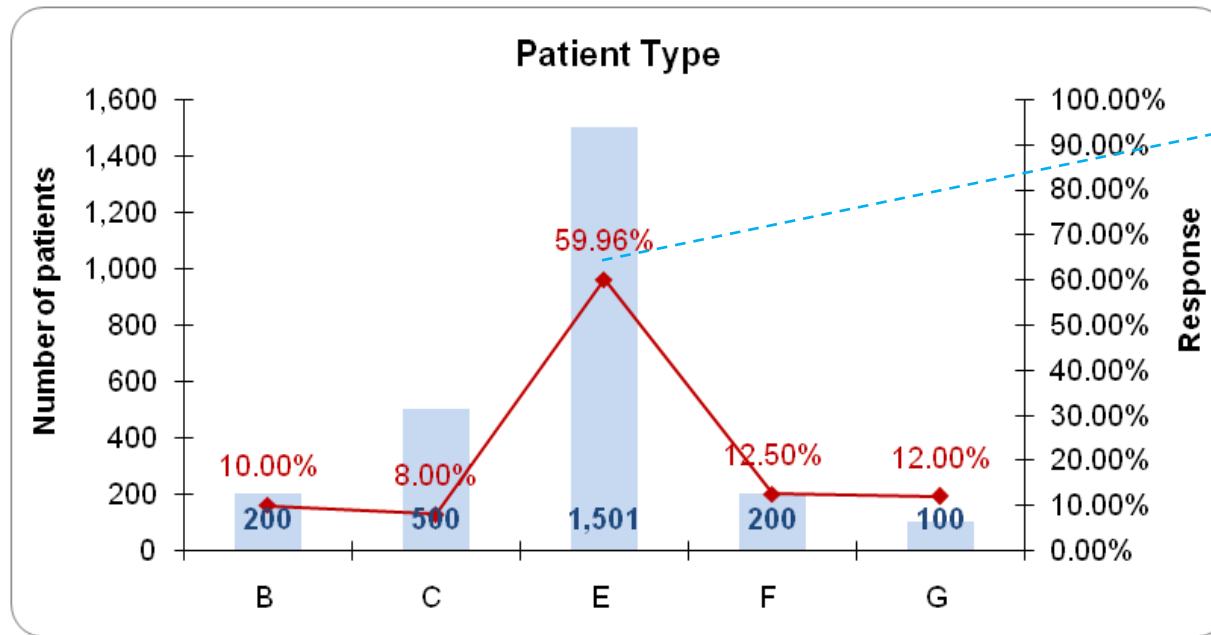
- Capping: Remove values above 1%tile / 99%tile
 - Check if those values actually make sense
- Remove Fictitious values
- Values beyond + / - 3 std. dev. of mean can be capped
- Inter Quartile Range (IQR):
 - $IQR = Q3 - Q1$
 - Points beyond the quartiles by 1.5 times IQR are deemed as potential outliers

Derived Variables

❖ Common transformations:

- Log transformations
 - Not applied to zero / negative values
 - Salary, policy sum assured – such values vary a lot based on individual and log transformations make them comparable
- Ratios
 - $\text{Price} = \text{Value} / \text{Volume}$;
 - $\text{Limit utilized on credit card} = \text{Spend} / \text{Total limit on card}$
- Conditional grouping: salary as High / Medium / Low
- Standardization:
 - Values having different units can be converted to scale having mean 0 and variance 1
- Reciprocals (NA for zero values)
- Linear transformations
- Raising to the power (Square, Cube, etc.)

Indicator variables



High response rate
for
one category

Patient type E
indicator
variable

1, if patient type
is E

0,
otherwise

Thank you !!!

“TORTURE THE
DATA, AND IT
WILL CONFESS
TO ANYTHING.”

– RONALD COASE, ECONOMICS, NOBEL PRIZE LAUREATE